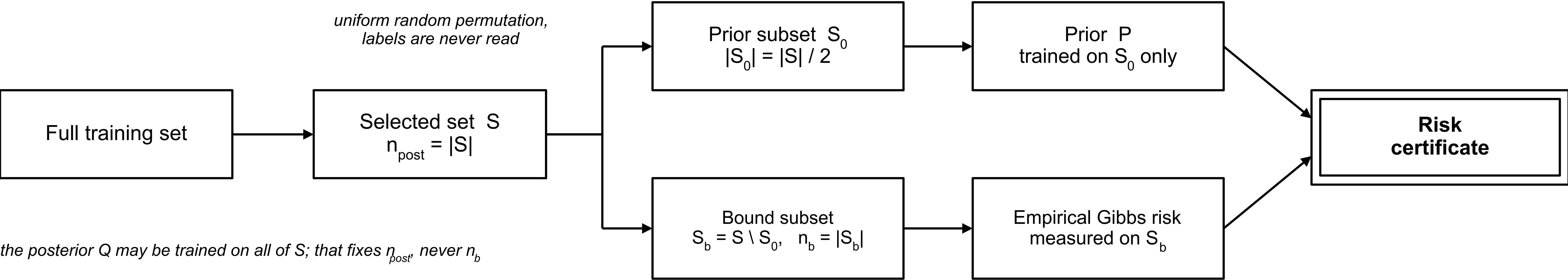
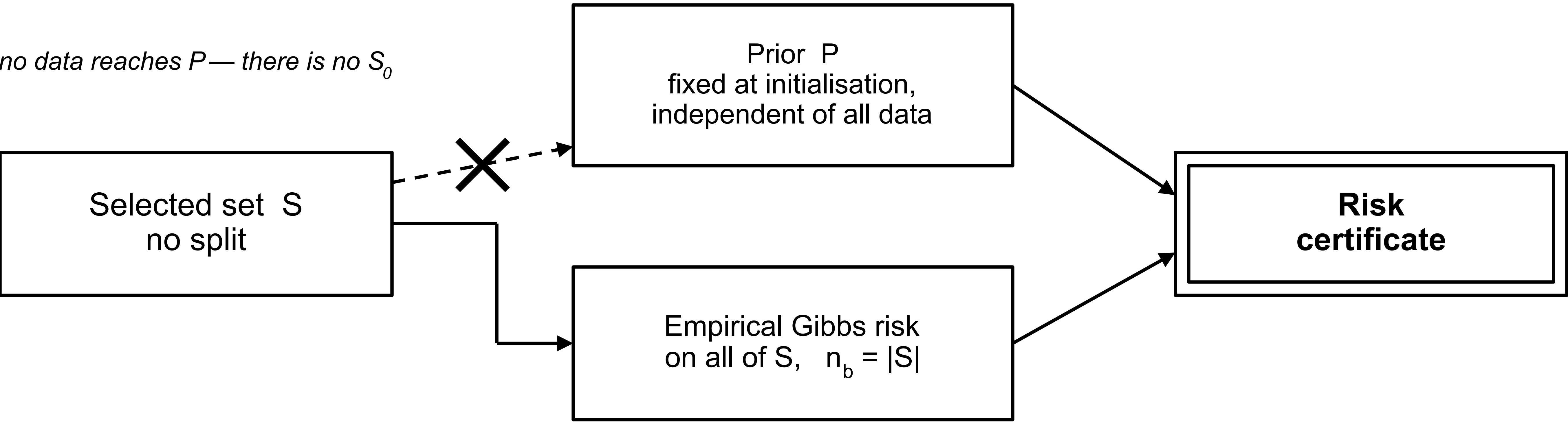


Case 1 — learnt (data-dependent) prior



Case 2 — data-free (random) prior



Worked counts, full data

Case 1 (even split)

MNIST: $n_{\text{post}} = 60,000, \quad n_b = 30,000$

CIFAR-10: $n_{\text{post}} = 50,000, \quad n_b = 25,000$

Putting 60,000 in the denominator here understates the complexity term and returns a certificate that is too tight.

Case 2 (no split)

MNIST: $n_b = |S| = 60,000$

CIFAR-10: $n_b = |S| = 50,000$

double border = reported quantity

n_b is the denominator of the bound

in Case 1, $S_0 \cap S_b = \emptyset$ and $S_0 \cup S_b = S$

dashed with $\times$ = no dependence at all